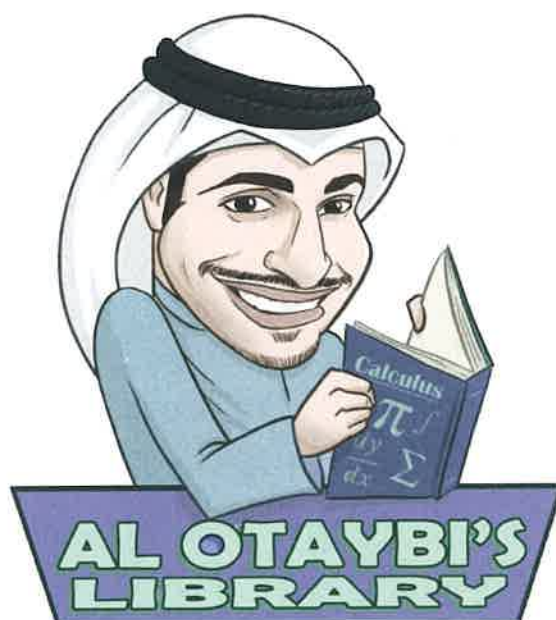


Kuwait University

2021/2022

Fall course



Calculus A
Final

Sections : 4.9 (Part b) – 5.2 (Part b) – 5.4

4.9 Antiderivatives.

5.2 The Definite Integral.

5.4 Indefinite Integrals and the Net Change Theorem .

(18)

$\int e^x dx$	$e^x + c$
$\int a^x dx$	$\frac{a^x}{\ln a} + c$



Big Mistakes



$$\int e^x dx = \frac{e^{x+1}}{x+1} + c$$

$$\int 5^x dx = \frac{5^{x+1}}{x+1} + c$$

Evaluate the following integral

$$\int e^{3x} dx = \frac{e^{3x}}{3} + c$$

$$\int 5^x dx = \frac{5^x}{\ln 5} + c$$

$$\int 5^{7x} dx = \frac{5^{7x}}{7 \ln 5} + c$$

$\int \frac{1}{x} dx$	$\ln x + c$
$\int \frac{1}{\sqrt{1-x^2}} dx$	$\sin^{-1}x + c$
$\int \frac{1}{x\sqrt{x^2-1}} dx$	$\sec^{-1}x + c$
$\int \frac{1}{x^2+1} dx$	$\tan^{-1}x + c$



Big Mistakes



$$\int \frac{1}{x} dx$$

~~$$= \int x^{-1} dx = \frac{x^0}{0} + c$$~~

$$\int \frac{1}{x^2} dx$$

~~$$= \ln|x^2| + c$$~~

Evaluate the following integral

$$\int \left(\frac{1}{x} + \frac{1}{x^2} - \frac{1}{\sqrt{x}} \right) dx$$

$$\begin{aligned} & \int \left(\frac{1}{x} + x^{-2} - x^{-\frac{1}{2}} \right) dx \\ &= \ln|x| + \frac{x^{-1}}{-1} - \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c \end{aligned}$$

Evaluate the following integral

$$\int \left(\frac{1}{t^2 + 1} + \frac{2}{\sqrt{1 - t^2}} \right) dt$$

$$= \tan^{-1} t + 2 \sin^{-1} t + c$$

$$\int \left(\frac{1}{5} - \frac{2}{x} \right) dx$$

$$= \frac{1}{5} x - 2 \ln|x| + c$$

$$\int \left(\frac{x^5 - x^3 + 2x}{x^4} + e^x \right) dx$$

$$= \int \left(\frac{x^5}{x^4} - \frac{x^3}{x^4} + \frac{2x}{x^4} + e^x \right) dx$$

$$= \int \left(x - \frac{1}{x} + \frac{2}{x^3} + e^x \right) dx$$

$$= \int \left(x - \frac{1}{x} + 2x^{-3} + e^x \right) dx$$

$$= \frac{x^2}{2} - \ln|x| + 2 \frac{x^{-2}}{-2} + e^x + c$$

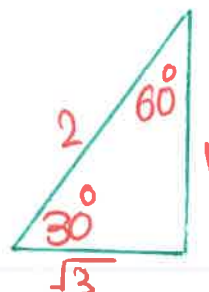
$$\frac{\sqrt{3}}{2}$$

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{1 - t^2}} dt$$

$$= \left[\sin^{-1} t \right]_0^{\frac{\sqrt{3}}{2}}$$

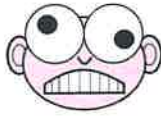
$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{3} - 0 = \frac{\pi}{3}$$





Text book question



Tricky question !!



$$\int \frac{2x^2 + 5}{x^2 + 1} dx$$

$$\begin{array}{r} 2x^2 + 5 \\ \underline{-(2x^2 + 2)} \\ 3 \end{array}$$
$$\int \left(2 + \frac{3}{x^2 + 1} \right) dx$$
$$= 2x + 3 \tan^{-1} x + C$$

Find $f(x)$ if

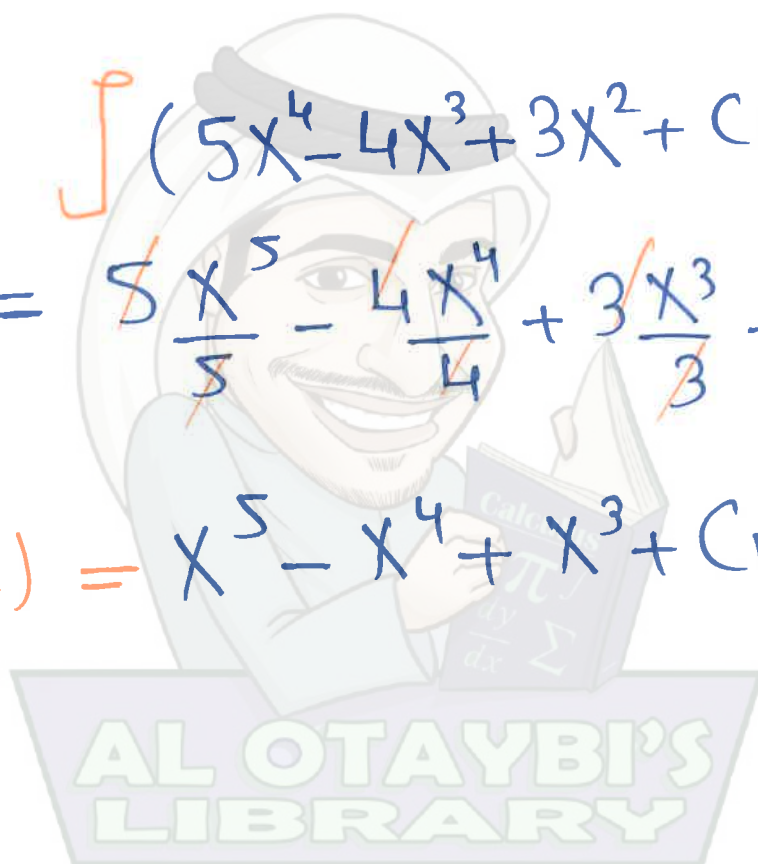
$$f''(x) = 20x^3 - 12x^2 + 6x$$

$$\begin{aligned} f'(x) &= \int (20x^3 - 12x^2 + 6x) \, dx \\ &= 20 \frac{x^4}{4} - 12 \frac{x^3}{3} + 6 \frac{x^2}{2} + C_1 \end{aligned}$$

$$\Rightarrow f'(x) = 5x^4 - 4x^3 + 3x^2 + C_1$$

$$\begin{aligned} f(x) &= \int (5x^4 - 4x^3 + 3x^2 + C_1) \, dx \\ &= \cancel{5} \frac{x^{\cancel{5}}}{\cancel{5}} - \cancel{4} \frac{x^{\cancel{4}}}{\cancel{4}} + \cancel{3} \frac{x^{\cancel{3}}}{\cancel{3}} + C_1 x + C_2 \end{aligned}$$

$$\Rightarrow f(x) = x^5 - x^4 + x^3 + C_1 x + C_2$$





Text book question



Find f if $f'(x) = e^x + 20(1+x^2)^{-1}$ and $f(0) = -2$

$$f(x) = \int (e^x + 20(1+x^2)^{-1}) dx$$

$$f(x) = \int \left(e^x + \frac{20}{1+x^2} \right) dx$$

$$f(x) = e^x + 20 \tan^{-1}x + c$$

$$f(0) = -2$$

$$f(0) = e^0 + 20 \tan^{-1}0 + c$$

$$-2 = 1 + 0 + c$$

$$-2 = 1 + c \quad \rightarrow \quad c = -3$$

$$\therefore f(x) = e^x + 20 \tan^{-1}x - 3$$

Find $f(x)$ if

$$f''(x) = \frac{3}{\sqrt{x}} \quad \text{if} \quad f(4) = 20, \quad f'(4) = 7$$

$$f'(x) = \int \frac{3}{\sqrt{x}} dx = \int 3x^{-\frac{1}{2}} dx$$

$$= 3 \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$f'(x) = 6x^{\frac{1}{2}} + C$$

$$\Rightarrow f'(4) = 7$$

$$7 = 6(4)^{\frac{1}{2}} + C$$

$$7 = 6(2) + C$$

$$7 = 12 + C \Rightarrow C = -5$$

$$f'(x) = 6x^{\frac{1}{2}} - 5$$

$$\Rightarrow f(x) = \int (6x^{\frac{1}{2}} - 5) dx$$

$$= 6 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - 5x + C$$

$$= 12 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - 5x + C$$

$$f(x) = 4x^{\frac{3}{2}} - 5x + C$$

$$\Rightarrow f(4) = 20$$

$$20 = 4(4)^{\frac{3}{2}} - 5(4) + C$$

$$20 = 4(2^2)^{\frac{3}{2}} - 20 + C$$

$$20 = 4(8) - 20 + C \rightarrow 20 = 32 - 20 + C$$

$$20 = 12 + C$$

$$C = 8$$

$$\Rightarrow f(x) = 4x^{\frac{3}{2}} - 5x + 8$$

Find $f(t)$ if

$$f''(t) = 2e^t + 3\sin t \quad \text{if } f(0) = 0 \quad f(\pi) = 0$$

$$f'(t) = \int (2e^t + 3\sin t) dt$$

$$f'(t) = 2e^t - 3\cos t + C_1$$

$$f(t) = \int (2e^t - 3\cos t + C_1) dt$$

$$f(t) = 2e^t - 3\sin t + C_1 t + C_2$$

$$\Rightarrow f(t) = 2e^t - 3\sin t + C_1 t + C_2$$

$$f(0) = 0$$

$$0 = 2e^0 - 3\sin 0 + C_1 \cdot 0 + C_2$$

$$0 = 2(1) + C_2 \Rightarrow C_2 = -2$$

$$\Rightarrow f(t) = 2e^t - 3\sin t + C_1 t - 2$$

$$f(\pi) = 0$$

$$0 = 2e^\pi - 3\sin \pi + C_1 \pi - 2$$

$$0 = 2e^\pi + C_1 \pi - 2$$

$$2 - 2e^\pi = C_1 \pi \Rightarrow C_1 = \frac{2 - 2e^\pi}{\pi}$$

$$\Rightarrow f(t) = 2e^t - 3\sin t + \frac{2 - 2e^\pi}{\pi} t - 2$$



تكاملات الدوال المثلثية



$\int \sin x \, dx$	$-\cos x + c$
$\int \cos x \, dx$	$\sin x + c$
$\int \sec^2 x \, dx$	$\tan x + c$
$\int \csc^2 x \, dx$	$-\cot x + c$
$\int \sec x \tan x \, dx$	$\sec x + c$
$\int \csc x \cot x \, dx$	$-\csc x + c$

$$\tan^2 x = \sec^2 x - 1$$

$$\cot^2 x = \csc^2 x - 1$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\sin x = \frac{1}{\csc x}$$

$$\csc x = \frac{1}{\sin x}$$

$$\cos x = \frac{1}{\sec x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\tan x = \frac{1}{\cot x}$$

$$\cot x = \frac{1}{\tan x}$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

$$\int_0^{\frac{\pi}{4}} (4 + 4\tan^2 x) dx$$

$$= \int_0^{\frac{\pi}{4}} [4 + 4(\sec^2 x - 1)] dx$$

$$= \int_0^{\frac{\pi}{4}} [\cancel{4} + 4\sec^2 x - \cancel{4}] dx$$

$$= \int_0^{\frac{\pi}{4}} 4\sec^2 x dx$$

$$= 4 [\tan x]_0^{\frac{\pi}{4}} = 4 [1 - 0] = 4$$

$$\int \cos^2 x dx$$

$$\frac{1}{2} \int (1 + \cos 2x) dx$$

$$\frac{1}{2} \left[x + \frac{\sin 2x}{2} \right] + C$$

$$\int \sin^2 5x dx$$

$$\frac{1}{2} \int (1 - \cos 10x) dx$$

$$\frac{1}{2} \left[x - \frac{\sin 10x}{10} \right] + C$$

$$\int \frac{\sec 2t}{\cos 2t} dt$$

$$= \int \sec 2t \frac{1}{\cos 2t} dt$$

$$= \int \sec 2t \sec 2t dt$$

$$= \int \sec^2 2t dt$$

$$= \frac{\tan 2t}{2} + C$$

$$\int \frac{\csc 5t}{\sin 5t} dt$$

$$= \int \csc 5t \csc 5t dt$$

$$= \int \csc^2 5t dt$$

$$= -\frac{\cot 5t}{5} + C$$

$$\int \frac{\csc x \cdot \cos x}{\sin x} dx$$

$$\int \csc x \left(\frac{\cos x}{\sin x} \right) dx$$

$$\int \csc x \cot x dx$$

$$= -\csc x + C$$

$$\int \frac{\sec x \cdot \sin x}{\cos x} dx$$

$$\int \sec x \tan x dx$$

$$\sec x + C$$



$$\int \frac{1 + \sin x}{\cos^2 x} dx$$

$$= \int \left[\frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} \right] dx$$

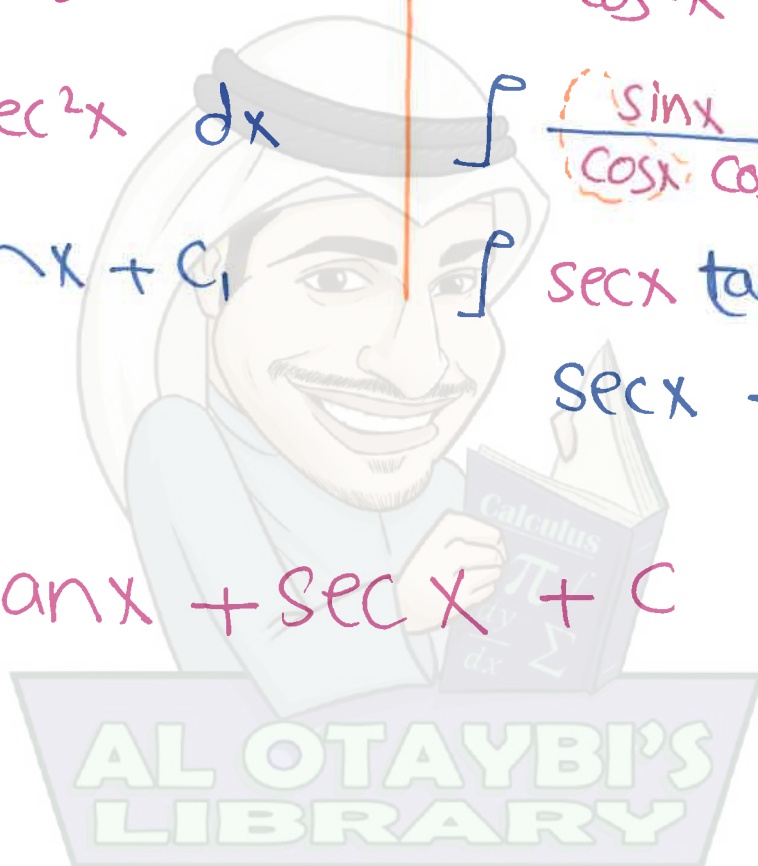
$$= \int \frac{1}{\cos^2 x} dx + \int \frac{\sin x}{\cos^2 x} dx$$

$$= \int \sec^2 x dx + \int \frac{\sin x}{\cos x \cos x} dx$$

$$= \tan x + C_1 + \int \sec x \tan x dx$$

$$\sec x + C_2$$

$$\Rightarrow \tan x + \sec x + C$$



$\int \sinh x \, dx$	$\cosh x + c$
$\int \cosh x \, dx$	$\sinh x + c$



Text book question



Find $f(x)$ if

$$f'(x) = x^3 + \sinh x, \quad f(0) = -2$$

$$f(x) = \int (x^3 + \sinh x) \, dx$$

$$f(x) = \frac{x^4}{4} + \cosh x + C_1$$

$$f(0) = -2$$

$$f(0) = 0 + \cosh 0 + C_1$$

$$-2 = 1 + C_1$$

$$C_1 = -3$$

$$\therefore f(x) = \frac{x^4}{4} + \cosh x - 3$$

تكاملات محددة استثنائية



$$\int_{-a}^a f(x) dx = 0$$

إذا كانت حدوده
(1) نفس العدد بإشارة مختلفة
(2) وكانت الدالة زوجية

$$2 \int_0^a f(x) dx$$

$$\int_{-a}^a f(x) dx = 0$$

إذا كانت حدوده
(1) نفس العدد بإشارة مختلفة
(2) وكانت الدالة فردية

ناتجه صفر

$$\int_a^a f(x) dx = 0$$

إذا كانت حدوده
نفس العدد
دائما وأبدا ناتجه صفر

$$\int_a^b |f(x)| dx = 0$$



عند وجود المطلق
يتم اعادة تعريفه بالنسبة لصفره

$$\int_{-a}^a \sqrt{a^2 - x^2} dx$$



إذا كانت حدوده
نفس العدد بإشارة مختلفة
احداها صفر
ربيع دائرة
نصف دائرة

$$\begin{aligned} y &= \sqrt{a^2 - x^2} \\ y^2 &= a^2 - x^2 \\ x^2 + y^2 &= a^2 \\ r^2 &= a^2 \\ r &= a \end{aligned}$$



Review



$$\int_{-3}^5 e \, dx$$

رقم

$$= e [x]_{-3}^5$$

$$= e [5 - (-3)]$$

$$= e [5 + 3] = 8e$$

$$\int_0^1 \left(x^e + e^x - \frac{1}{e} \right) dx$$

رقم

$$= \left[\frac{x^{e+1}}{e+1} + e^x - \frac{1}{e} x \right]_0^1$$

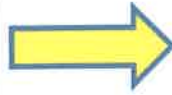
$$= \left[\left(\frac{1^{e+1}}{e+1} + e^1 - \frac{1}{e} \right) - \left(\frac{0}{e+1} + e^0 - 0 \right) \right]$$

$$= \frac{1}{e+1} + e - \frac{1}{e} - (0 + 1)$$

$$= \frac{1}{e+1} + e - \frac{1}{e} - 1$$



$$\int_a^a f(x) dx = 0$$



إذا كانت حدوده
نفس العدد
دائما وأبدا ناتجه صفر



Examples



$$\int_{-6}^{-6} (x^3 + 5x^2) dx$$

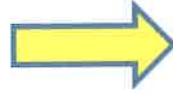
=

$$\int_1^1 \left(e^x + \frac{1}{x} \right) dx$$

=



$$\int_a^b |f(x)| dx = 0$$



عند وجود المطلق
يتم إعادة تعريفه بالنسبة لصفره



Examples



$$\int_{-6}^2 |x+2| dx$$

$$\int_{-6}^{-2} (x+2) dx + \int_{-2}^2 (x+2) dx \quad \begin{matrix} \xrightarrow{\quad} x+2=0 \\ x=-2 \\ \in (-6, 2) \end{matrix}$$

$$= \left[\frac{x^2}{2} + 2x \right]_{-6}^{-2} + \left[\frac{x^2}{2} + 2x \right]_{-2}^2$$

$$= \left[\left(\frac{4}{2} - 4 \right) - \left(\frac{36}{2} - 12 \right) \right] + \left[\left(\frac{4}{2} + 4 \right) - \left(\frac{4}{2} - 4 \right) \right]$$

$$= \left[(2 - 4) - (18 - 12) \right] + \left[(2 + 4) - (2 - 4) \right]$$

$$= \left[-2 - 6 \right] + \left[6 - (-2) \right]$$

$$= \left[-8 \right] + 8$$

$$8 + 8 = 16$$

$$\int_0^2 |x^2 - 1| dx$$

$$x^2 - 1 = 0$$

$$(x-1)(x+1)$$

$$x=1 \quad x=-1$$

$$\in \quad \notin$$

$$(0,2) \quad (0,2)$$

$$-\int_0^1 (x^2 - 1) dx + \int_1^2 (x^2 - 1) dx$$

$$- \left[\frac{x^3}{3} - x \right]_0^1 + \left[\frac{x^3}{3} - x \right]_1^2$$

$$- \left[\left(\frac{1}{3} - 1 \right) - 0 \right] + \left[\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right]$$

$$-\frac{1}{3} + 1 + \frac{8}{3} - 2 - \frac{1}{3} + 1$$

$$-\frac{1}{3} + \frac{8}{3} - \frac{1}{3} = \frac{6}{3} = 2$$

$$\int_0^2 |x^2 + x - 2| dx$$

$$x^2 + x - 2 = 0$$

$$(x-1)(x+2) = 0$$

$$x=1 \quad x=-2$$

$$\in \quad \notin$$

$$(0,2) \quad (0,2)$$

$$-\int_0^1 (x^2 + x - 2) dx + \int_1^2 (x^2 + x - 2) dx$$

$$- \left[\frac{x^3}{3} + \frac{x^2}{2} - 2x \right]_0^1 + \left[\frac{x^3}{3} + \frac{x^2}{2} - 2x \right]_1^2$$

$$- \left[\left(\frac{1}{3} + \frac{1}{2} - 2 \right) - 0 \right] + \left[\left(\frac{8}{3} + \frac{4}{2} - 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right) \right]$$

$$-\frac{1}{3} - \frac{1}{2} + 2 + \frac{8}{3} + 2 - 4 - \frac{1}{3} - \frac{1}{2} + 2$$

$$-\frac{1}{3} + \frac{8}{3} - \frac{1}{3} - \frac{1}{2} - \frac{1}{2} + 2 + 2 - 4 + 2$$

$$\frac{6}{3} - 1 + 2 = 2 - 1 + 2 = 3$$

$$\int_0^2 |x^2 - x^3| dx$$

$$\begin{aligned} x^2 - x^3 &= 0 \\ x^2(1-x) &= 0 \\ x=0 & \quad x=1 \\ \oplus & \quad \ominus \\ (0,2) & \quad (0,2) \end{aligned}$$

$$\int_0^1 (x^2 - x^3) dx + \ominus \int_1^2 (x^2 - x^3) dx$$

$$\left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 - \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_1^2$$

$$\left(\frac{1}{3} - \frac{1}{4} \right) - 0 - \left[\left(\frac{8}{3} - \frac{16}{4} \right) - \left(\frac{1}{3} - \frac{1}{4} \right) \right]$$

$$\frac{1}{3} - \frac{1}{4} - \left[\frac{8}{3} - 4 - \frac{1}{3} + \frac{1}{4} \right]$$

$$\frac{1}{3} - \frac{1}{4} - \frac{8}{3} + 4 + \frac{1}{3} - \frac{1}{4}$$

$$\frac{1}{3} - \frac{8}{3} + \frac{1}{3} - \frac{1}{4} - \frac{1}{4} + 4$$

$$-\frac{6}{3} - \frac{2}{4} + 4$$

$$-2 - \frac{1}{2} + 4$$

$$-\frac{1}{2} + 2$$

$$-\frac{1}{2} + \frac{4}{2} = \frac{3}{2}$$

$$\int_{-1}^2 (x+1)|x-1| dx$$

$$x-1=0$$

$$x=1$$

$$\in (-1, 2)$$

$$-\int_{-1}^1 (x+1)(x-1) dx + \int_1^2 (x+1)(x-1) dx$$

$$-\int_{-1}^1 (x^2-1) dx + \int_1^2 (x^2-1) dx$$

$$-\left[\frac{x^3}{3}-x\right]_{-1}^1 + \left[\frac{x^3}{3}-x\right]_1^2$$

$$-\left[\left(\frac{1}{3}-1\right)-\left(-\frac{1}{3}+1\right)\right] + \left[\left(\frac{8}{3}-2\right)-\left(\frac{1}{3}-1\right)\right]$$

$$-\left[\frac{1}{3}-1+\frac{1}{3}-1\right] + \left[\frac{8}{3}-2-\frac{1}{3}+1\right]$$

$$-\frac{1}{3}+1-\frac{1}{3}+1+\frac{8}{3}-2-\frac{1}{3}+1$$

$$-\frac{1}{3}-\frac{1}{3}+\frac{8}{3}-\frac{1}{3}+1$$

$$\frac{5}{3}+1$$

$$\frac{5}{3}+\frac{3}{3}=\frac{8}{3}$$

$$\int_1^4 \frac{|4-x^2|}{x^2} dx$$

$$4-x^2=0$$

$$(2-x)(2+x)=0$$

$$x=2 \quad x=-2$$

$$\in \quad \notin$$

$$(1,3) \quad (1,3)$$

$$\int_1^2 \frac{4-x^2}{x^2} dx + \ominus \int_2^4 \frac{4-x^2}{x^2} dx$$

$$\int_1^2 \left(\frac{4}{x^2} - \frac{x^2}{x^2} \right) dx - \int_2^4 \left(\frac{4}{x^2} - \frac{x^2}{x^2} \right) dx$$

$$\int_1^2 (4x^{-2} - 1) dx - \int_2^4 (4x^{-2} - 1) dx$$

$$\left[\frac{4x^{-1}}{-1} - x \right]_1^2 - \left[\frac{4x^{-1}}{-1} - x \right]_2^4$$

$$\left[-\frac{4}{x} - x \right]_1^2 - \left[-\frac{4}{x} - x \right]_2^4$$

$$\left[\left(-\frac{4}{2} - 2 \right) - \left(-\frac{4}{1} - 1 \right) \right] - \left[\left(-\frac{4}{4} - 4 \right) - \left(-\frac{4}{2} - 2 \right) \right]$$

$$-2 - 2 - (-5) - \left[(-1 - 4) - (-2 - 2) \right]$$

$$-4 + 5 - \left[-5 - (-4) \right]$$

$$1 - \left[-5 + 4 \right]$$

$$1 - \left[-1 \right] = 1 + 1 = 2$$

$$\int_0^{\pi} |\cos x| dx$$

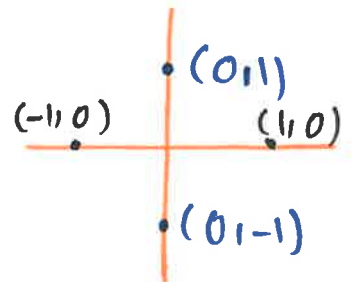
$$\cos x = 0$$

$$\int_0^{\frac{\pi}{2}} \cos x dx + \int_{\frac{\pi}{2}}^{\pi} -\cos x dx$$

$$[\sin x]_0^{\frac{\pi}{2}} - [\sin x]_{\frac{\pi}{2}}^{\pi}$$

$$[1 - 0] - [0 - 1]$$

$$1 + 1 = 2$$



$$x = \frac{\pi}{2} \in (0, \pi)$$

$$x = \frac{3\pi}{2} \notin (0, \pi)$$



$$\int_{-2}^2 |x^2 - 1| dx$$



$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$x = 1 \quad x = -1$$

$$\in (-2, 2) \quad \in (-2, 2)$$

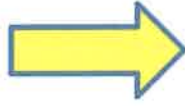
$$\int_{-2}^{-1} (x^2 - 1) dx + \int_{-1}^1 -(x^2 - 1) dx + \int_1^2 (x^2 - 1) dx$$

$$\left[\frac{x^3}{3} - x \right]_{-2}^{-1} - \left[\frac{x^3}{3} - x \right]_{-1}^1 + \left[\frac{x^3}{3} - x \right]_1^2$$

تم بحمد الله تعالى



$$\int_{-a}^a f(x) dx = 0$$



إذا كانت حدوده
 (1) نفس العدد بإشارة مختلفة
 (2) وكانت الدالة فردية
 ناتجه صفر



Examples

$$\int_{-1}^1 \frac{x}{\sqrt{9-x^2}} dx$$

Let $f(x) = \frac{x}{\sqrt{9-x^2}}$

$$f(-x) = \frac{(-x)}{\sqrt{9-(-x)^2}}$$

$$= \frac{-x}{\sqrt{9-x^2}} = -f(x)$$

∴ $f(x)$ is odd function

$$\therefore \int_{-1}^1 \frac{x}{\sqrt{9-x^2}} dx = 0$$

$$\int_{-2}^2 \frac{\sin x}{\sqrt{4-x^2}} dx$$

Let $f(x) = \frac{\sin x}{\sqrt{4-x^2}}$

$$f(-x) = \frac{\sin(-x)}{\sqrt{4-(-x)^2}} = \frac{-\sin x}{\sqrt{4-x^2}} = -f(x)$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-2}^2 \frac{\sin x}{\sqrt{4-x^2}} dx = 0$$

$$\int_{-5}^5 x^2 \tan x dx$$

Let $f(x) = x^2 \tan x$

$$f(-x) = (-x)^2 \tan(-x)$$

$$= x^2 [-\tan x]$$

$$= -x^2 \tan x = -f(x)$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-5}^5 f(x) dx = 0$$

$$\int_{-e}^e \frac{\cos x}{x} dx$$

Let $f(x) = \frac{\cos x}{x}$

$$f(-x) = \frac{\cos(-x)}{-x} = \frac{\cos x}{-x} = -f(x)$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-e}^e \frac{\cos x}{x} dx = 0$$

$$\int_{-1}^1 x^3 \sqrt{x^4 + 1} dx$$

Let $f(x) = x^3 \sqrt{x^4 + 1}$

$$f(-x) = (-x)^3 \sqrt{(-x)^4 + 1}$$

$$= -x^3 \sqrt{x^4 + 1} = -f(x)$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-1}^1 x^3 \sqrt{x^4 + 1} dx = 0$$

$$\int_{-4}^4 \frac{2+x^2}{x^4 \sin x} dx$$

Let $f(x) = \frac{2+x^2}{x^4 \sin x}$

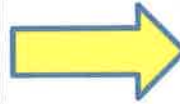
$$\begin{aligned} f(-x) &= \frac{2+(-x)^2}{(-x)^4 \sin(-x)} \\ &= \frac{2+x^2}{x^4 [-\sin x]} \\ &= \frac{2+x^2}{-x^4 \sin x} = -f(x) \end{aligned}$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-4}^4 \frac{2+x^2}{x^4 \sin x} dx = 0$$



$$\int_{-a}^a \sqrt{a^2 - x^2} dx = 0$$



إذا كانت حدوده
نفس العدد بإشارة مختلفة ← نصف دائرة
أحداها صفر ← ربع دائرة



Examples

$$\int_0^2 \sqrt{4 - x^2} dx$$

$$y = \sqrt{a^2 - x^2}$$

$$y^2 = a^2 - x^2$$

$$x^2 + y^2 = a^2$$

$$r^2 = a^2$$

$$r = a$$

$$y = \sqrt{4 - x^2}$$

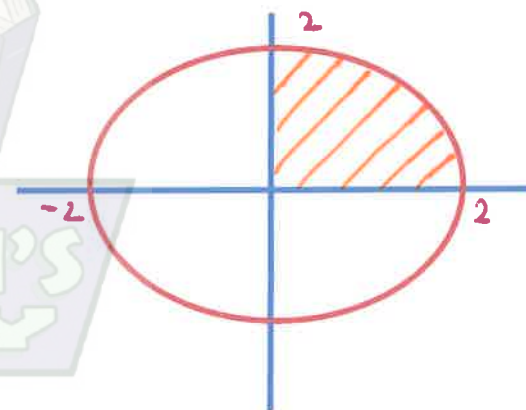
$$y^2 = 4 - x^2$$

$$x^2 + y^2 = 4$$

$$r^2 = 4$$

$$r = 2$$

$$\begin{aligned} \Rightarrow A &= \frac{1}{4} \pi r^2 \\ &= \frac{1}{4} \pi (2)^2 \\ &= \frac{1}{4} \pi (4) \\ &= \pi \end{aligned}$$



$$\int_{-3}^3 \sqrt{9 - x^2} dx$$

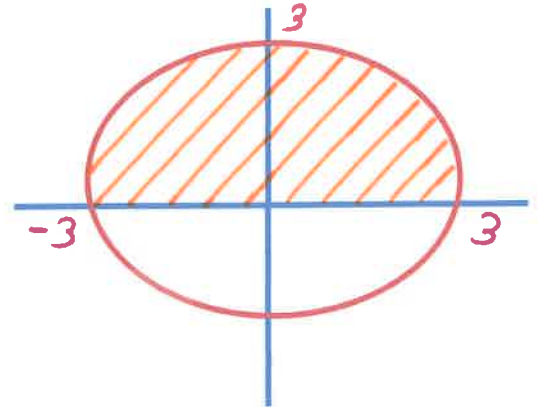
$$y = \sqrt{9 - x^2}$$

$$y^2 = 9 - x^2$$

$$x^2 + y^2 = 9$$

$$r^2 = 9 \rightarrow r = 3$$

$$\begin{aligned} \Rightarrow A &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi (3)^2 \\ &= \frac{1}{2} \pi (9) = \frac{9}{2} \pi \end{aligned}$$



$$\int_{-e}^e \sqrt{e^2 - x^2} dx$$

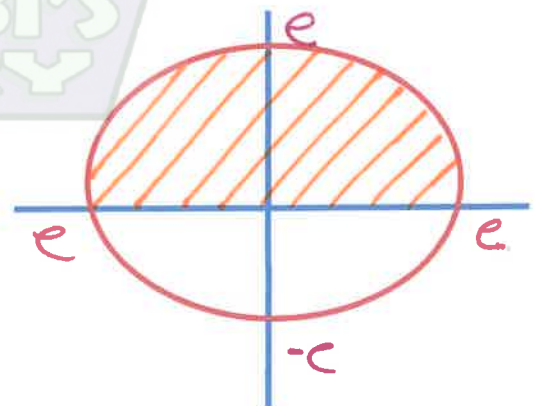
$$y = \sqrt{e^2 - x^2}$$

$$y^2 = e^2 - x^2$$

$$x^2 + y^2 = e^2$$

$$r^2 = e^2 \rightarrow r = e$$

$$\begin{aligned} \Rightarrow A &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi e^2 \end{aligned}$$



$$\int_{-\pi}^{\pi} (x^3 - \sqrt{\pi^2 - x^2}) dx$$

$$\int_{-\pi}^{\pi} x^3 dx - \int_{-\pi}^{\pi} \sqrt{\pi^2 - x^2} dx$$

Let $f(x) = x^3$

$$f(-x) = (-x)^3$$

$$= -x^3$$

$$= -f(x)$$

$\Rightarrow \therefore f(x)$ is odd function

$$\therefore \int_{-\pi}^{\pi} x^3 dx = 0$$

$$y = \sqrt{\pi^2 - x^2}$$

$$y^2 = \pi^2 - x^2$$

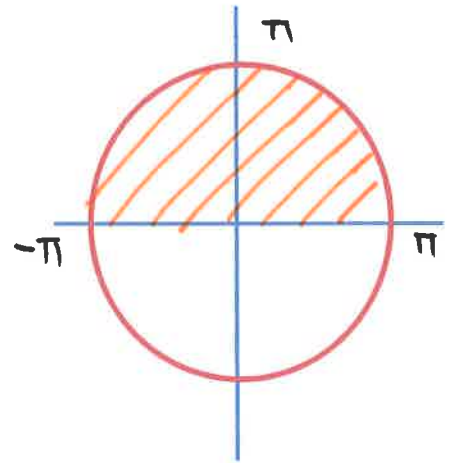
$$x^2 + y^2 = \pi^2$$

$$r^2 = \pi^2 \rightarrow r = \pi$$

$$A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi (\pi)^2$$

$$= \frac{1}{2} \pi^3$$



$$- \frac{1}{2} \pi^3$$

$$= -\frac{1}{2} \pi^3$$

$$\int_{-2}^2 \sqrt{16 - 4x^2} dx$$

$$\int_{-2}^2 \sqrt{4(4 - x^2)} dx$$

$$\int_{-2}^2 \sqrt{4} \sqrt{4 - x^2} dx$$

$$2 \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$y = \sqrt{4 - x^2}$$

$$y^2 = 4 - x^2$$

$$x^2 + y^2 = 4$$

$$r^2 = 4 \rightarrow r = 2$$



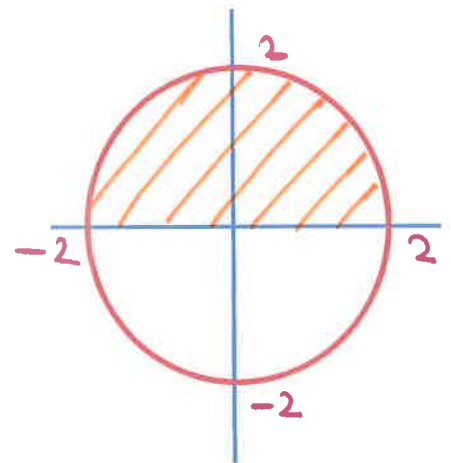
$$A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi (2)^2$$

$$= \frac{1}{2} \pi (4)$$

$$= 2\pi$$

$$2 \cdot 2\pi = 4\pi$$



$$\int_{-3}^3 (x-3)\sqrt{9-x^2} dx$$

ما بهير لاهول دائرة

هناك x قبل الجذر

$$\int_{-3}^3 (x\sqrt{9-x^2} - 3\sqrt{9-x^2}) dx$$

$$\int_{-3}^3 x\sqrt{9-x^2} dx - \int_{-3}^3 3\sqrt{9-x^2} dx$$

Let $f(x) = x\sqrt{9-x^2}$

$$\begin{aligned} f(-x) &= (-x)\sqrt{9-(-x)^2} \\ &= -x\sqrt{9-x^2} \\ &= -f(x) \end{aligned}$$

$\Rightarrow \therefore f(x)$ is odd function

$$\therefore \int_{-3}^3 x\sqrt{9-x^2} dx = 0$$

$$3 \int_{-3}^3 \sqrt{9-x^2} dx$$

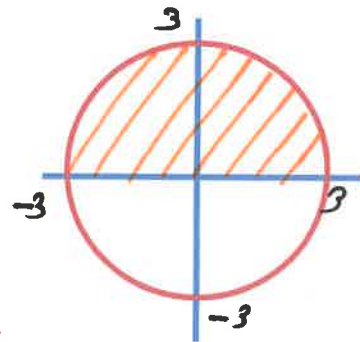
$$y = \sqrt{9-x^2}$$

$$y^2 = 9-x^2$$

$$x^2 + y^2 = 9$$

$$r^2 = 9$$

$$r = 3$$



$$\Rightarrow A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi (3)^2$$

$$= \frac{1}{2} \pi 9 = \frac{9}{2} \pi$$

$$\Rightarrow 0 - 3 \frac{9}{2} \pi = -\frac{27}{2} \pi$$

$$\int_{-1}^1 (1-x)\sqrt{1-x^2} dx$$

ما هي دالتك
هناك x
قبل
البذر

$$\int_{-1}^1 (\sqrt{1-x^2} - x\sqrt{1-x^2}) dx$$

$$\int_{-1}^1 \sqrt{1-x^2} dx - \int_{-1}^1 x\sqrt{1-x^2} dx$$

$$y = \sqrt{1-x^2}$$

$$y^2 = 1-x^2$$

$$x^2 + y^2 = 1$$

$$r^2 = 1$$

$$r = 1$$

$$f(x) = x\sqrt{1-x^2}$$

$$f(-x) = (-x)\sqrt{1-(-x)^2}$$

$$= -x\sqrt{1-x^2}$$

$$= -f(x)$$

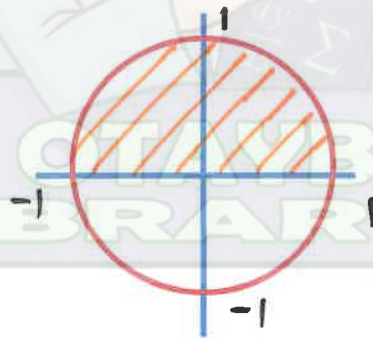
$\therefore f(x)$ is odd

$$\therefore \int_{-1}^1 f(x) dx = 0$$

$$\Rightarrow A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi (1)^2$$

$$= \frac{1}{2} \pi$$



$$\Rightarrow \frac{1}{2} \pi - 0 = \frac{1}{2} \pi$$

$$\int_{-3}^3 (\sqrt{9-x^2} + x^2 \sin^3 x + x) dx$$

$$\int_{-3}^3 \sqrt{9-x^2} dx$$

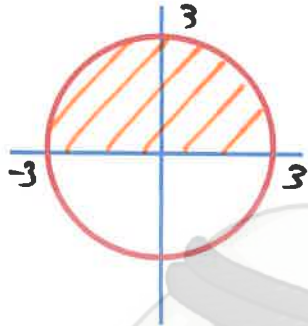
$$+ \int_{-3}^3 x^2 \sin^3 x dx + \int_{-3}^3 x dx$$

$$y = \sqrt{9-x^2}$$

$$y^2 = 9-x^2$$

$$x^2 + y^2 = 9$$

$$r^2 = 9 \rightarrow r = 3$$



$$\begin{aligned} \Rightarrow A &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi (3)^2 \\ &= \frac{1}{2} \pi 9 \\ &= \frac{9}{2} \pi \end{aligned}$$

$$f(x) = x^2 \sin^3 x = x^2 [\sin x]^3$$

$$\begin{aligned} f(-x) &= (-x)^2 [\sin(-x)]^3 \\ &= x^2 [-\sin x]^3 \\ &= -x^2 (\sin x)^3 \\ &= -f(x) \end{aligned}$$

$\therefore f(x)$ is odd

$$\therefore \int_{-3}^3 x^2 \sin^3 x dx = 0$$

$$f(x) = x$$

$$\begin{aligned} f(-x) &= -x \\ &= -f(x) \end{aligned}$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-3}^3 x dx = 0$$

AL OTAYBI'S LIBRARY

$$\int_{-1}^1 x^2 \tan x \, dx =$$

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a) 0 b) 1 c) -1 d) 2 e) None of the above

$$f(x) = x^2 \tan x$$

$$f(-x) = (-x)^2 \tan(-x) = x^2 \tan(-x) = -x^2 \tan(x) = -f(x)$$

∴ $f(x)$ is odd function

$$\therefore \int_{-1}^1 f(x) \, dx = 0$$

Evaluate each of the following integrals

$$\int_{-1}^1 (x^2 + \cos x) \tan x \, dx$$

Dec 2017

$$\int_{-1}^1 x^2 \tan x \, dx + \int_{-1}^1 \cos x \tan x \, dx$$

$$f(x) = x^2 \tan x$$

$$\begin{aligned} f(-x) &= (-x)^2 \tan(-x) \\ &= x^2 \tan(-x) \\ &= -x^2 \tan(x) \\ &= -f(x) \end{aligned}$$

∴ $f(x)$ is odd function

$$\therefore \int_{-1}^1 f(x) \, dx = 0$$

$$g(x) = \cos x \tan x$$

$$\begin{aligned} g(-x) &= \cos(-x) \tan(-x) \\ &= -\cos x \tan x \\ &= -g(x) \end{aligned}$$

∴ $g(x)$ is odd function

$$\therefore \int_{-1}^1 g(x) \, dx = 0$$

$$\therefore \int_{-1}^1 (x^2 + \cos x) \tan x \, dx = 0$$



Properties of the Definite Integral



$$\int_b^a f(x) dx = - \int_a^b f(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$



Comparison Properties of the Integral

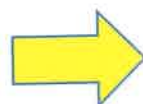


if $f(x) \geq 0$ for $a \leq x \leq b$



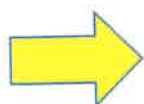
$$\int_a^b f(x) dx \geq 0$$

if $f(x) \geq g(x)$ for $a \leq x \leq b$



$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

if $m \leq f(x) \leq M$ for $a \leq x \leq b$



$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$$



Text book question



$$\text{If } \int_0^9 f(x) dx = 37 \text{ and } \int_0^9 g(x) dx = 16, \text{ find}$$

$$\int_0^9 [2f(x) + 3g(x)] dx$$

$$\int_0^9 [2f(x) + 3g(x)] dx$$

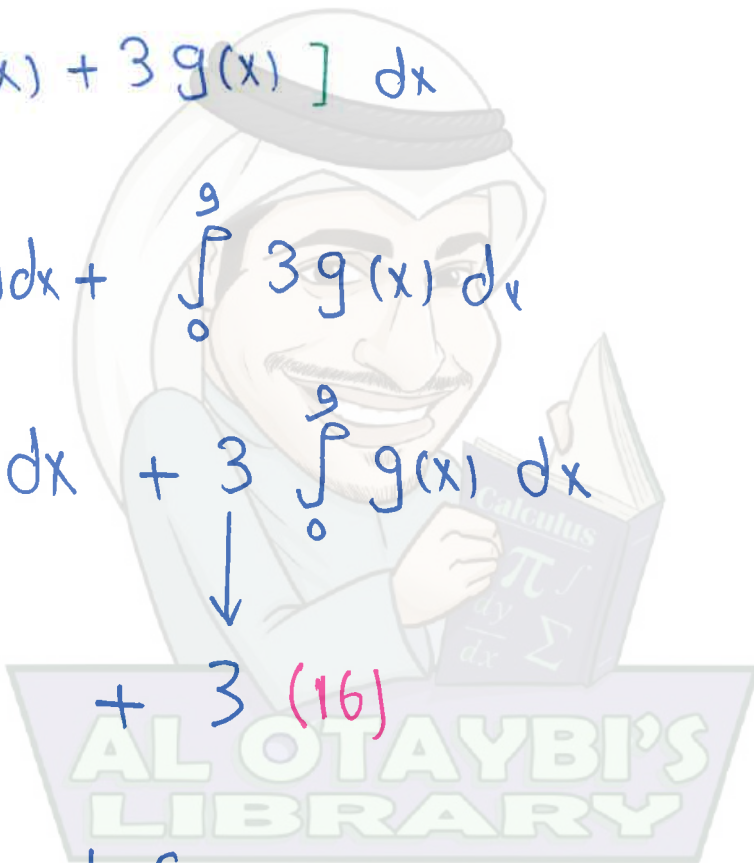
$$\int_0^9 2f(x) dx + \int_0^9 3g(x) dx$$

$$2 \int_0^9 f(x) dx + 3 \int_0^9 g(x) dx$$

$$\downarrow \qquad \qquad \downarrow$$
$$2(37) + 3(16)$$

$$= 74 + 48$$

$$= 122$$

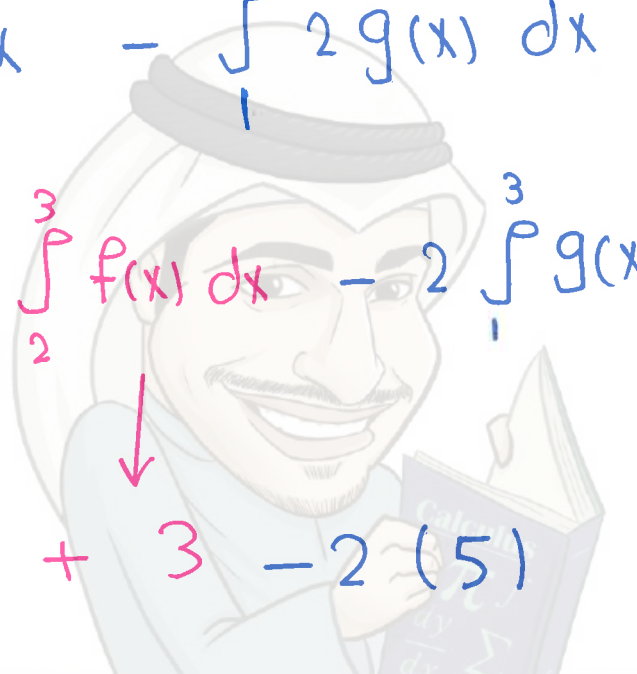


Let f and g be two continuous functions such that

$$\int_2^1 f(x) dx = -1, \quad \int_2^3 f(x) dx = 3 \quad \text{and} \quad \int_1^3 g(x) dx = 5$$

Find $\int_1^3 [f(x) - 2g(x)] dx$

May 2017


$$\begin{aligned} & \int_1^3 f(x) dx - \int_1^3 2g(x) dx \\ & \int_1^2 f(x) dx + \int_2^3 f(x) dx - 2 \int_1^3 g(x) dx \\ & \downarrow \qquad \qquad \downarrow \\ & - \int_2^1 f(x) dx + 3 - 2(5) \\ & -(-1) + 3 - 10 \\ & 1 + 3 - 10 = -6 \end{aligned}$$



Text book questions



If $\int_2^8 f(x) dx = 7.3$ and $\int_2^4 f(x) dx = 5.9$

Find $\int_4^8 f(x) dx$

$$\begin{aligned}
 & \int_4^8 f(x) dx \\
 &= \int_4^2 f(x) dx + \int_2^8 f(x) dx \\
 &= - \int_2^4 f(x) dx + 7.3 \\
 &= - 5.9 + 7.3 = 1.4
 \end{aligned}$$

Write as a single integral in the form $\int_a^b f(x) dx$:

$$\int_{-2}^2 f(x) dx + \int_2^5 f(x) dx - \int_{-2}^{-1} f(x) dx$$

$$\int_{-2}^2 f(x) dx + \int_2^5 f(x) dx - \int_{-2}^{-1} f(x) dx$$

$$\int_{-2}^5 f(x) dx - \int_{-2}^{-1} f(x) dx$$

$$\int_{-2}^5 f(x) dx + \int_{-1}^2 f(x) dx$$

$$\int_{-1}^2 f(x) dx + \int_{-2}^5 f(x) dx = \int_{-1}^5 f(x) dx$$



Text book questions



Find $\int_0^5 f(x) dx$ if

$$f(x) = \begin{cases} 3 & \text{for } x < 3 \\ x & \text{for } x \geq 3 \end{cases}$$

$$\int_0^5 f(x) dx$$

$$= \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

$$= \int_0^3 3 dx + \int_3^5 x dx$$

$$= [3x]_0^3 + \left[\frac{x^2}{2} \right]_3^5$$

$$= 9 - 0 + \frac{25}{2} - \frac{9}{2}$$

$$= 9 + \frac{16}{2}$$

$$= 9 + 8$$

$$= 17$$

Show that problems

$$\int f(x) dx = g(x)$$

تکامل غیر محدد = ناتج



يتم اشتقاق الطرفين
وإثبات تساويهما

Note

$$\frac{d}{dx} \int f(x) dx = f(x)$$

$$\dots \leq \int_a^b f(x) dx \leq \dots$$

تکامل على شكل متباينة



وضع علاقة بين الطرفين ثم
يتم العمل على هذه العلاقة

$$\int_a^b f(x) dx = \int_c^d g(x) dx$$

تکامل محدد = تکامل محدد



توصيل أحد التكاملين للآخر
or
حل التكاملين وإثبات تساوي النواتج

Show that :

$$\int \sin^3 x \sqrt{\cos x} dx = \frac{2}{7} (\cos x)^{\frac{7}{2}} - \frac{2}{3} (\cos x)^{\frac{3}{2}} + c$$

Differentiate L.H.S

$$\frac{d}{dx} \int \sin^3 x \sqrt{\cos x} dx = \sin^3 x \sqrt{\cos x} \Rightarrow \textcircled{1}$$

Differentiate R.H.S

$$\begin{aligned} & \frac{d}{dx} \left[\frac{2}{7} (\cos x)^{\frac{7}{2}} - \frac{2}{3} (\cos x)^{\frac{3}{2}} + c \right] \\ &= \frac{2}{7} \cdot \frac{7}{2} (\cos x)^{\frac{7}{2}-1} (-\sin x) - \frac{2}{3} \cdot \frac{3}{2} (\cos x)^{\frac{3}{2}-1} (-\sin x) + 0 \\ &= -\sin x (\cos x)^{\frac{5}{2}} + \sin x (\cos x)^{\frac{1}{2}} \\ &= \sin x (\cos x)^{\frac{1}{2}} \left[-(\cos x)^2 + 1 \right] \\ &= \sin x (\cos x)^{\frac{1}{2}} \left[-(\cos^2 x) + 1 \right] \\ & \quad \swarrow \begin{aligned} & \cos^2 x + \sin^2 x = 1 \\ & \sin^2 x = 1 - \cos^2 x \end{aligned} \\ &= \sin x (\cos x)^{\frac{1}{2}} \sin^2 x \\ &= \sin^3 x (\cos x)^{\frac{1}{2}} \Rightarrow \textcircled{2} \end{aligned}$$

From $\textcircled{1}$ $\textcircled{2} \Rightarrow \therefore \underline{\text{L.H.S}} = \underline{\text{R.H.S}}$

Show that :

$$\int \sin^3(x) dx = -\frac{1}{3} \sin^2(x) \cdot \cos(x) - \frac{2}{3} \cos(x) + c$$

Differentiate L.H.S

$$\frac{d}{dx} \int \sin^3 x dx = \sin^3 x \implies (1)$$

Differentiate R.H.S

$$\frac{d}{dx} \left[-\frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x + c \right]$$

استعمل قاعدة بـسـا + بسـا استعمل

$$= -\frac{1}{3} 2 \sin x \cos x \cos x + \left(-\frac{1}{3} \sin^2 x \right) (-\sin x) + \frac{2}{3} \sin x + 0$$

$$= -\frac{2}{3} \sin x \cos^2 x + \frac{1}{3} \sin^3 x + \frac{2}{3} \sin x$$

$$= -\frac{2}{3} \sin x [1 - \sin^2 x] + \frac{1}{3} \sin^3 x + \frac{2}{3} \sin x$$

$$= -\frac{2}{3} \sin x + \frac{2}{3} \sin^3 x + \frac{1}{3} \sin^3 x + \frac{2}{3} \sin x$$

$$= \frac{3}{3} \sin^3 x = \sin^3 x \implies (2)$$

From (1) (2) $\implies \therefore$ L.H.S = R.H.S

Show that :

$$\int \sin^3 x \, dx = \frac{1}{3} \cos^3 x - \cos(x) + c$$

Where c is constant

Differentiate L.H.S

$$\frac{d}{dx} \int \sin^3 x \, dx = \sin^3 x \implies (1)$$

Differentiate R.H.S

$$\begin{aligned} \frac{d}{dx} \left[\frac{1}{3} \cos^3 x - \cos x + c \right] \\ \frac{1}{3} 3 \cos^2 x (-\sin x) + \sin x + 0 \\ -\sin x \cos^2 x + \sin x \end{aligned}$$

Sol #1

Sol #2

$$\begin{aligned} &= -\sin x [1 - \sin^2 x] + \sin x && \sin x [-\cos^2 x + 1] \\ &= -\sin x + \sin^3 x + \sin x && \sin x \sin^2 x \\ &= \sin^3 x \implies (2) && = \sin^3 x \implies (3) \end{aligned}$$

From (1) (2) $\implies$ L.H.S = R.H.S

Let $F(x) = x \ln x + x + c$, where c is a constant,
and $f(x) = 2 + \ln x$.

Show that $F(x)$ is the most general anti derivative of $f(x)$

$$\int (2 + \ln x) dx = x \ln x + x + c$$

بأنما الي فيها c
هيا ناتج للتكامل

Differentiate L.H.S

$$\frac{d}{dx} \int (2 + \ln x) dx = 2 + \ln x \Rightarrow \textcircled{1}$$

Differentiate R.H.S

$$\frac{d}{dx} [x \ln x + x + c]$$

اشتقا بشتا + بشتا اشتقا

$$(1) \ln x + x \cdot \frac{1}{x} + 1 + 0$$

$$\ln x + 1 + 1 = \ln x + 2 \Rightarrow \textcircled{2}$$

$\Rightarrow$ From $\textcircled{1} \textcircled{2} \rightarrow \therefore \underline{\text{L.H.S}} = \underline{\text{R.H.S}}$

Show that $F(x) = \frac{1}{2}(x^2 \tan^{-1} x - x + \tan^{-1} x)$ is the antiderivative of $f(x) = x \tan^{-1} x$

July 2018

$$\int x \tan^{-1} x \, dx = \frac{1}{2}(x^2 \tan^{-1} x - x + \tan^{-1} x)$$

Differentiate L.H.S

$$\frac{d}{dx} \int x \tan^{-1} x \, dx = x \tan^{-1} x \longrightarrow (1)$$

Differentiate R.H.S

$$\frac{d}{dx} \left[\frac{1}{2}(x^2 \tan^{-1} x - x + \tan^{-1} x) \right]$$

$$\frac{1}{2} \left(2x \tan^{-1} x + x^2 \frac{1}{x^2+1} - 1 + \frac{1}{x^2+1} \right)$$

$$\frac{1}{2} \left(2x \tan^{-1} x + \frac{x^2+1}{x^2+1} - 1 \right)$$

$$\frac{1}{2} (2x \tan^{-1} x + 1 - 1)$$

$$\frac{1}{2} (2x \tan^{-1} x) = x \tan^{-1} x \longrightarrow (2)$$

From (1) (2)

$\therefore F(x)$ is the antiderivative of $f(x)$.

Let $f(x) = \sin(\ln x)$. Show that

Dec 2017

$F(x) = \frac{x}{2} [\sin(\ln x) - \cos(\ln x)]$ is the antiderivative of $f(x)$

$$\int \sin(\ln x) dx = \frac{x}{2} [\sin(\ln x) - \cos(\ln x)]$$

Differentiate L.H.S

$$\frac{d}{dx} \int \sin(\ln x) dx = \sin(\ln x) \rightarrow (1)$$

Differentiate R.H.S

$$\frac{d}{dx} \frac{x}{2} [\sin(\ln x) - \cos(\ln x)]$$

اشتق + مشتق + اشتق

$$= \frac{1}{2} [\sin(\ln x) - \cos(\ln x)] + \frac{x}{2} [\cos(\ln x) \frac{1}{x} + \sin(\ln x) \frac{1}{x}]$$

$$= \frac{1}{2} \sin(\ln x) - \frac{1}{2} \cos(\ln x) + \frac{1}{2} \cos(\ln x) + \frac{1}{2} \sin(\ln x)$$

$$= \frac{2}{2} \sin(\ln x) = \sin(\ln x) \rightarrow (2)$$

From (1) (2)

* $F(x)$ is the antiderivative of $f(x)$.

Show that $2 \leq \int_0^1 (3 - \cos^5(7x+1)) \leq 4$

May 2018

$$-1 \leq \cos^5 \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \cos^5(7x+1) \leq 1$$

$$\times -1$$

$$+1 \quad \parallel \quad -\cos^5(7x+1) \quad \parallel \quad -1$$

$$+3$$

$$4 \quad \parallel \quad 3 - \cos^5(7x+1) \quad \parallel \quad 2$$

$$\int_0^1 4 \, dx \quad \parallel \quad \int_0^1 (3 - \cos^5(7x+1)) \, dx \quad \parallel \quad \int_0^1 2 \, dx$$



$$4 [x]_0^1$$

$$4 [1-0]$$

$$2 [x]_0^1$$

$$2 [1-0]$$

$$4 \quad \parallel \quad \int_0^1 (3 - \cos^5(7x+1)) \, dx \quad \parallel \quad 2$$

Prove that :

$$\int_{-6}^{12} (8 - \cos^{10} x) dx \geq 0$$

$$0 \leq \cos^{10} x \leq 1 \quad \forall x$$

* -1

$$0 \leq 1 - \cos^{10} x \leq 1$$

+8

$$8 \leq 8 - \cos^{10} x \leq 7$$

$$8 - \cos^{10} x \leq 7$$

$$\int_{-6}^{12} (8 - \cos^{10} x) dx \leq \int_{-6}^{12} 7 dx$$

$$7 [x]_{-6}^{12}$$

$$7 [12 - (-6)]$$

$$7 [18]$$

$$126$$

$$\Rightarrow \therefore \int_{-6}^{12} (8 - \cos^{10} x) dx \leq 126 \leq 0$$

$$\therefore \int_{-6}^{12} (8 - \cos^{10} x) dx \leq 0$$

Show that on any interval $[a, b]$ we have

$$\int_a^b \frac{\sqrt{5+7x^2}}{3-\cos x} dx \geq \int_a^b \frac{\sqrt{5+7x^2}}{4} dx$$

$$-1 \leq \cos x \leq 1$$

$$\underline{\underline{* -1}}$$

$$1 \geq -\cos x \geq -1$$

$$\underline{\underline{+3}}$$

$$4 \geq 3 - \cos x \geq 2$$

$$\frac{1}{4} \leq \frac{1}{3 - \cos x} \leq \frac{1}{2}$$

$$\frac{1}{4} \leq \frac{1}{3 - \cos x}$$

$$* \sqrt{5+7x^2}$$

$$\frac{\sqrt{5+7x^2}}{4}$$

$$\frac{\sqrt{5+7x^2}}{3 - \cos x}$$

$$\int_a^b \frac{\sqrt{5+7x^2}}{4} dx \leq \int_a^b \frac{\sqrt{5+7x^2}}{3 - \cos x} dx$$

Prove that :

$$\frac{3}{4} \leq \int_{\frac{1}{2}}^2 \frac{1}{x} dx \leq 3$$

$$\frac{1}{2} \leq x \leq 2$$

$$\forall x \in [\frac{1}{2}, 2]$$

$$2 \geq \frac{1}{x} \geq \frac{1}{2}$$

$$\int_{\frac{1}{2}}^2 2 dx$$

$$\geq \int_{\frac{1}{2}}^2 \frac{1}{x} dx$$

$$\geq \int_{\frac{1}{2}}^2 \frac{1}{x} dx$$

$$\geq \int_{\frac{1}{2}}^2 \frac{1}{2} dx$$

$$\int_{\frac{1}{2}}^2 \frac{1}{2} dx$$

$$\geq \int_{\frac{1}{2}}^2 \frac{1}{2} dx$$

$$2 [x]_{\frac{1}{2}}^2$$

$$\frac{1}{2} [x]_{\frac{1}{2}}^2$$

$$2 [2 - \frac{1}{2}]$$

$$\frac{1}{2} [2 - \frac{1}{2}]$$

$$2 [\frac{4}{2} - \frac{1}{2}]$$

$$\frac{1}{2} [\frac{4}{2} - \frac{1}{2}]$$

$$2 \cdot \frac{3}{2}$$

$$\frac{1}{2} \cdot \frac{3}{2}$$

$$3$$

$$\frac{3}{4}$$

$$\Rightarrow 3 \geq \int_{\frac{1}{2}}^2 \frac{1}{x} dx \geq \frac{3}{4}$$

Prove that :

$$2 \leq \int_{-1}^1 \sqrt{1+x^2} dx \leq 2\sqrt{2}$$

+1 $0 \leq x^2 \leq 1 \quad \forall x \in [-1, 1]$

$1 \leq 1+x^2 \leq 2$

$\sqrt{1} \leq \sqrt{1+x^2} \leq \sqrt{2}$

$1 \leq \sqrt{1+x^2} \leq \sqrt{2}$

$\int_{-1}^1 (1) dx \leq \int_{-1}^1 \sqrt{1+x^2} dx \leq \int_{-1}^1 \sqrt{2} dx$

$[x]_{-1}^1$

$1 - (-1)$
2

$\sqrt{2} [x]_{-1}^1$

$\sqrt{2} [1 - (-1)]$
 $\sqrt{2} \cdot 2$

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$\Rightarrow 2 \leq \int_{-1}^1 \sqrt{1+x^2} dx \leq 2\sqrt{2}$

Show that :

$$48 \leq \int_0^3 (x^2 + 16) dx \leq 75$$

$$0 \leq x^2 \leq 3^2 \quad \forall x \in [0, 3]$$

$$0 \leq x^2 \leq 9$$

$$+16$$

$$16 \leq x^2 + 16 \leq 25$$

$$\int_0^3 16 dx \leq \int_0^3 (x^2 + 16) dx \leq \int_0^3 25 dx$$

$$16 [x]_0^3$$

$$16 [3 - 0]$$

$$48$$

$$25 [x]_0^3$$

$$25 [3 - 0]$$

$$75$$

$$\Rightarrow 48 \leq \int_0^3 (x^2 + 16) dx \leq 75$$

Show that :

$$\int_0^3 (x^2 + 2)^4 dx \geq 48$$

$$0 \leq x^2 \leq 3^2 \quad \forall x \in [0, 3]$$

$$0 \leq x^2 \leq 9$$

$$+ 2$$

$$2 \leq x^2 + 2 \leq 11$$

$$(2)^4 \leq (x^2 + 2)^4 \leq (11)^4$$

$$(2)^4 \leq (x^2 + 2)^4$$

$$16 \leq (x^2 + 2)^4$$

$$\int_0^3$$

$$16 dx \leq \int_0^3 (x^2 + 2)^4 dx$$

$$16 [x]_0^3$$

$$16 [3 - 0]$$

$$48$$

$$\Rightarrow \therefore 48 \leq \int_0^3 (x^2 + 2)^4 dx$$

Show that if $m \leq xf(x) \leq M$ for $0 < a \leq x \leq b$ then

$$m \ln\left(\frac{b}{a}\right) \leq \int_a^b f(x) dx \leq M \ln\left(\frac{b}{a}\right)$$

$$m \leq xf(x) \leq M$$

$\div x$

$$\frac{m}{x} \leq f(x) \leq \frac{M}{x}$$

$$\int_a^b \frac{m}{x} dx \leq \int_a^b f(x) dx \leq \int_a^b \frac{M}{x} dx$$

$$m [\ln|x|]_a^b \leq M [\ln|x|]_a^b$$

$$m [\ln b - \ln a] \leq M [\ln b - \ln a]$$

$$m \ln\left(\frac{b}{a}\right) \leq M \ln\left(\frac{b}{a}\right)$$

$$\Rightarrow m \ln\left(\frac{b}{a}\right) \leq \int_a^b f(x) dx \leq M \ln\left(\frac{b}{a}\right)$$

Prove that :

$$\int_1^4 \cos^2 x \, dx = 3 - \int_1^4 \sin^2 x \, dx$$

SOL #1

$$\text{L.H.S} = \int_1^4 \cos^2 x \, dx$$

$$= \int_1^4 (1 - \sin^2 x) \, dx$$

$$= \int_1^4 (1) \, dx - \int_1^4 \sin^2 x \, dx$$

$$[x]_1^4$$

$$[4-1]$$

$$3$$

$$- \int_1^4 \sin^2 x \, dx$$

$$= \text{R.H.S}$$

Prove that :

$$\int_1^4 \cos^2 x \, dx = 3 - \int_1^4 \sin^2 x \, dx$$

Sol #2

$$\text{R.H.S} = 3 - \int_1^4 \sin^2 x \, dx$$

$$= 3 - \int_1^4 (1 - \cos^2 x) \, dx$$

$$= 3 - \left[\int_1^4 (1) \, dx - \int_1^4 \cos^2 x \, dx \right]$$

$$\downarrow$$
$$[x]_1^4$$

$$\frac{4-1}{3}$$

$$= 3 - \left[3 - \int_1^4 \cos^2 x \, dx \right]$$

$$= \cancel{3} - \cancel{3} + \int_1^4 \cos^2 x \, dx$$

$$= \int_1^4 \cos^2 x \, dx = \text{L.H.S}$$

Prove that :

$$\int_1^4 \cos^2 x \, dx = 3 - \int_1^4 \sin^2 x \, dx$$

Sol #3

$$\int_1^4 \cos^2 x \, dx + \int_1^4 \sin^2 x \, dx = 3$$

$$\text{L.H.S} = \int_1^4 \cos^2 x \, dx + \int_1^4 \sin^2 x \, dx$$

$$= \int_1^4 (\cos^2 x + \sin^2 x) \, dx$$

$$= \int_1^4 (1) \, dx$$

$$= [x]_1^4 = 4 - 1 = 3$$

$$= \text{R.H.S}$$